

ORF 524 – Midterm Exam

Matias D. Cattaneo

Department of Operations Research and Financial Engineering

Princeton University

October 23, 2019

Contents

1	Question 1: Probability Theory (15 points)	1
2	Question 2: Finite-Sample Statistical Inference (35 points)	3

Notes:

- Duration of examination: *3 hours*.
- All parts are equally weighted (5 points).
- Please start each question on a separate page.
- Please provide as much detail as possible in your answers.
- Answers without proper justification will not receive (partial) credit.

1 Question 1: Probability Theory (15 points)

1. Let (X, Y) be a bivariate random vector on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose that $X \sim \text{Bernoulli}(p)$ and $Y \sim \text{Bernoulli}(q)$ for some $p, q \in (0, 1)$. Show the following:

$$X \perp\!\!\!\perp Y \quad \Longleftrightarrow \quad \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$

Show that the “if” direction is false in general. Hint: take $X \sim \mathcal{N}(0, 1)$ and $Y = X^2$.

2. Let (X, Y, Z) be a square-integrable random vector on $(\Omega, \mathcal{F}, \mathbb{P})$. Solve the following problem:

$$\min_{g \in \mathcal{G}} \mathbb{E} [(\mathbb{E}[Y|X] - g(Z))^2], \quad \text{where } \mathcal{G} = \{g : \mathbb{E}[|g(Z)|^2] < \infty\}.$$

In words, find the best mean square error approximation to the conditional expectation of Y given X based on square-integrable functions of Z .

3. Let $(X_i : 1 \leq i \leq n)$ be i.i.d. copies of the random variable X on $(\Omega, \mathcal{F}, \mathbb{P})$ and with well-defined Moment Generating Function $M_X(t)$. Using the Markov inequality, show that for all x and $t > 0$:

$$\mathbb{P}[\bar{X}_n \geq x] \leq \frac{\prod_{i=1}^n M_{X_i}(t/n)}{\exp(tx)}, \quad \text{where } \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Find the sharpest upper bound for each x .

2 Question 2: Finite-Sample Statistical Inference (35 points)

Assume $(X_i : 1 \leq i \leq n)$ is a random sample from $X \sim \mathbb{P}_\theta$ with Lebesgue density

$$f_X(x; \theta) = g(\theta) \mathbb{1}(a \leq x \leq \theta),$$

where $\theta \in \Theta = (a, \infty)$, $a \geq 0$ known, and $g : \Theta \mapsto \mathbb{R}$ is a fixed function.

1. Show that $g(\theta) = (\theta - a)^{-1}$, and find the c.d.f. of X , denoted $F_X(x)$.
2. Show that $X_{(n)} = \max_{1 \leq i \leq n} X_i$ is a complete sufficient statistic for θ .
3. Show that $\mathbb{E}[X_{(n)}]$ is downward biased. Construct a UMVU estimator of θ .
4. Consider the following hypothesis testing problem:

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta \neq \theta_0$$

- (a) Show that $T_n(\theta_0) = (X_{(n)} - a)/(\theta_0 - a)$ is a pivotal statistic (under H_0).
Identify its probability law.
- (b) Propose a $100(1 - \alpha)\%$ confidence interval for θ , where $\alpha \in [0, 1]$.
- (c) Construct an α -level testing procedure based on the pivot $T_n(\theta_0)$.
- (d) Derive the power function, denoted $\beta_n(\theta)$, and sketch a plot of it.